GLOBAL ENERGY HACKATHON 2026 · INNOVATIVE & AFFORDABLE SOLUTIONS

Kesho AI

Forecasts a village's power demand from satellite roofs and open data, then right-sizes and finances its solar mini-grid, for developers, lenders and electrification agencies



BRIEF
03 Predict mini-grid demand before building

TRACK
Digital Solutions and AI

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IMPACT · 25% OF SCORE

Villages get a mini-grid sized right the first time

THE PROBLEM

490 million people need mini-grids by 2030: 217,000 grids, about $127 billion. At today's pace only about 44,800 get built, and each one is sized from guesses.

"Measured use runs 5 to 35 kWh per customer a month, and forecasts are often overstated." (ESMAP 2022)

2,888

households connected by year 5 across 6 villages in 6 countries, each with a P10 to P90 demand range

37% → 15%

chance of brownouts by year 5: one P50 build vs Kesho AI's staged plan, with far less idle kit

$1.1M

saved at Gbamu Gbamu, Nigeria by sharing one plant with 14 settlements within 5 km

INNOVATION · 25% OF SCORE

Demand, sizing and finance from open data

TODAY

Old census data and surveys asking people to guess use of power they have never had. One system is built for year 5, so it sits idle early and browns out later.

OUR APPROACH

Count every roof from space, run 4,000 demand futures checked against metered sites within 1,000 miles, and build in stages that keep brownout odds under 15%.

ROOT CAUSE WE ADDRESS

No demand data before building, so sizing risk lands on the village and the lender.

HOW IT WORKS

1. Count and classify every roof from Google, Microsoft and OpenStreetMap, checked with fresh Sentinel-2 images
2. Forecast 5 years of demand with ranges, tested on 10 years of real weather and climate outlooks
3. Right-size in stages, price the wires or grid line, and find the grant or loan that pays back

SUSTAINABILITY & AFFORDABILITY · 25% OF SCORE

It pays back without ongoing grants

COST TO THE USER

$5 to $8 / month

household bill at $0.85 a kWh, year 5, against $5 to $20 families spend on kerosene, candles and batteries

PAYBACK

4 to 6 years

WHO PAYS

Households pay per kWh; results-based grants (about $433 a connection) and concessional loans cover the build. Kesho AI finds the smallest grant needed.

HOW IT PAYS FOR ITSELF

Bills cover running costs and 1.2x the loan payment from year 2. Each stage is added only when meters pass 80% of what is installed.

WHO KEEPS IT RUNNING

The operator's local technicians; battery and inverter renewal around year 11 is in the 20-year plan.

ENVIRONMENTAL BENEFIT

About 290 MWh a year of solar at Gbamu Gbamu (230 kWp) instead of kerosene and diesel generators.

A working prototype on six real villages



EVIDENCE IT WORKS

- Gbamu Gbamu's real 85 kWp plant matches our year-2 demand (84 kWp)
- Open data: checked on 2,225 tagged buildings
- Risk: no meters yet.
- Fix: one-day household count

PATH TO SCALE

PILOT

Gbamu Gbamu, Nigeria: 794 customers, builds on the existing 85 kWp plant

EXPAND

Villages within 10 km, and REA and SEforALL sites in Nigeria and Sierra Leone

SCALE

Any village in Africa: data covers the continent; bundling cuts cost per customer

OUR ASK

A pilot developer, meter data from 50 to 200 sites, and an introduction to REA Nigeria

REA DARES (run by Nigeria's Rural Electrification Agency) and SEforALL's Universal Energy Facility are grant programmes that pay mini-grid developers for each home they connect.